

ZHENGRONG WANG

Room 931, Ho Sin-Hang Engineering Building, CUHK, Shating, N.T., Hong Kong SAR, PRC

zhengrongwang@cuhk.edu.hk <https://seanzw.github.io> Google Scholar

EXPERIENCE

Assistant Professor , <i>Department of Computer Science and Engineering</i> The Chinese University of Hong Kong	Hong Kong SAR, China Aug. 25th 2025 - Now
Research Scientist HiSilicon	Shenzhen, China Aug. 25th 2024 - Aug. 24th 2025
Postdoctoral Scholar , <i>Department of Computer Science</i> University of California, Los Angeles	Los Angeles, USA Jan. 1st 2024 - Aug. 24th. 2024

EDUCATION

University of California, Los Angeles , <i>Department of Computer Science</i> Ph.D. in Computer Science, Advisor: Tony Nowatzki	Los Angeles, USA Aug. 2018 - Dec. 2023
University of California, Los Angeles , <i>Department of Computer Science</i> Master of Science in Computer Science Thesis: An LLVM-IR Datagraph-Based Simulator for Flexible Design Space Exploration over Accelerator Architectures	Los Angeles, USA Sep. 2016 - Jul. 2018
Tsinghua University , <i>Department of Electronic Engineering</i> Bachelor of Engineering in Electronic Engineering, GPA: 91/100 Thesis: Optimizing Convolutional Neural Network on FPGA under Heterogeneous Computing Framework with OpenCL	Beijing, China Aug. 2012 - Jul. 2016

PUBLICATION - TOP 5 MOST REPRESENTATIVE WITHIN 5 YEARS

Affinity Alloc: Taming Not-So Near-Data Computing Zhengrong Wang , Christopher Liu, Nathan Beckmann, Tony Nowatzki <i>IEEE/ACM International Symposium on Microarchitecture (MICRO)</i> , 2023, Toronto, Canada.
Infinity Stream: Portable and Programmer-Friendly In-/Near-Memory Fusion Zhengrong Wang , Christopher Liu, Aman Arora, Lizy John, Tony Nowatzki <i>ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)</i> , 2023, Vancouver, Canada.
Near-Stream Computing: General and Transparent Near-Cache Acceleration Zhengrong Wang , Jian Weng, Sihao Liu, Tony Nowatzki <i>IEEE International Symposium on High-Performance Computer Architecture (HPCA)</i> , 2022, Seoul, South Korea.
Stream Floating: Enabling Proactive and Decentralized Cache Optimizations Zhengrong Wang , Jian Weng, Jason Lowe-Power, Jayesh Gaur, Tony Nowatzki <i>IEEE International Symposium on High-Performance Computer Architecture (HPCA)</i> , 2021, Seoul, South Korea. Best Paper Runner-Up
Stream-Based Memory Access Specialization for General Purpose Processors Zhengrong Wang , Tony Nowatzki <i>ACM International Symposium on Computer Architecture (ISCA)</i> , 2019, Phoenix, USA.

PUBLICATION - 5 REPRESENTATIVE BEYOND RECENT 5 YEARS

PIMSAB: A Processing-In-Memory System with Spatially-Aware Communication and Bit-Serial-Aware Computation Siyuan Ma, Kaustubh Mhatre, Jian Weng, Bagus Hanindhito, Zhengrong Wang (5th/8) , Tony Nowatzki, Lizy John, Aman Arora <i>2024 ACM Transactions on Architecture and Code Optimization (TACO)</i> .
SPGPU: Spatially Programmed GPU Shizhuo Zhu, Illia Shkirko, Jacob Levinson, Zhengrong Wang (4th/5) , Tony Nowatzki <i>2024 IEEE Computer Architecture Letters (CAL)</i> .

OverGen: Improving FPGA Usability through Domain-specific Overlay Generation
 Sihao Liu, Jian Weng, Dylan Kupsh, Atefeh Sohrabizadeh, **Zhengrong Wang (5th/11)**, Licheng Guo, Jiuyang Liu,
 Maxim Zhulin, Lucheng Zhang, Jason Cong, Tony Nowatzki
IEEE/ACM International Symposium on Microarchitecture (MICRO), 2022, Chicago, USA.

Best Paper Runner-Up

DSAGEN: Synthesizing Programmable Spatial Accelerators
 Jian Weng, Sihao Liu, Vidushi Dadu, **Zhengrong Wang (4th/6)**, Preyas Shah, Tony Nowatzki
ACM International Symposium on Computer Architecture (ISCA), 2020, virtual.

IEEE Micro Top Picks Honorable Mention

A Hybrid Systolic-Dataflow Architecture for Inductive Matrix Algorithms
 Jian Weng, Sihao Liu, **Zhengrong Wang (3rd/5)**, Vidush Dadu, Tony Nowatzki
IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2020, San Diego, USA.

AWARDS AND HONORS

Dissertation Year Fellowship, <i>UCLA</i>	<i>June. 2023</i>
Best Paper Runner-Up (OverGen, in <i>MICRO '22</i>), <i>IEEE</i>	<i>Oct. 2022</i>
Best Paper Runner-Up (Stream Floating, in <i>HPCA '21</i>), <i>IEEE</i>	<i>Feb. 2021</i>
IEEE Micro Top Picks 2020 Honorable Mention (DSAGEN, in <i>ISCA '20</i>), <i>IEEE</i>	<i>Jan. 2021</i>
2021 Dongguan Entrepreneur Scholarship, <i>Dongguan Entrepreneurs Federation</i>	<i>Nov. 2021</i>
Second-class Scholarship for Excellent Freshmen, <i>Tsinghua University</i>	<i>Oct. 2012</i>
Wang Zhaosheng Scholarship for Excellent Student from Dongguan, <i>Wang Zhaosheng Foundation</i>	<i>Oct. 2012</i>

OPEN SOURCE PROJECTS & INFRASTRUCTURES

Stream-Specialized Near-Data Acceleration Framework 	<i>Jan. 2018 - Now</i>
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First Author & Maintainer

- Full-stack implementation of stream-specialized near-data acceleration.
- Include LLVM-based compiler transformation and end-to-end simulation in gem5.
- Results published in ISCA' 19, HPCA' 21, HPCA '22 and ASPLOS '23. More in submission.

Gem5-AVX 	<i>Jan. 2019 - Now</i>
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First Author & Maintainer

- Add AVX-512 support to gem5 simulator, extensively used in research.
- Faithfully model the microarchitecture of vectorized instructions, including microops.
- Detailed tutorials on how to support new instructions.

Gem5 Simulator 	<i>Jan. 2019 - Now</i>
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Committer & Maintainer

- Contribute various bug fixes for instruction decoding, microarchitecture deadlock.
- Review pull requests.